IFAC
INTERNATIONAL FEDERATION OF AUTOMATIC CONTROL



6th IFAC Conference on Engine and Powertrain Control, Simulation and Modeling, 23-25 August, 2021, Tokyo, Japan

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Further information please visit:
http://shenlab.jp/ecosm2021

Welcome to E-COSM 2021

E-COSM was created in 2006 as a triennial workshop series. After four editions, it was updated as an IFAC conference series since 2018. The conference is sponsored by the IFAC Technical Committee on Automotive Control (TC7.1). Following the past successful workshops and conference, E-COSM 2021 will move to Tokyo and will be organized by Sophia University, Tokyo. E-COSM 2021 aims to bring researchers and practitioners from industry and academia together to present and discuss the latest research progresses, as well as industrial experience in deploying "real-world" applications. Accepted papers except for discussion papers and two page industrial papers, will be published in IFAC-PapersOnLine in partnership with Elsevier.

News

Due to COVID-19, the organizing committee has decided to host E-COSM 2021 as an online virtual conference.

Topics

Authors from academia and industry are encouraged to join the event and share their research results, E-COSM 2021 topics include, but not limited to:

1. System modeling and simulation frameworks for engine and powertrain control design, validation and calibration.
2. Model-based control, estimation and diagnostics for the new generations of CI and SI engines and their exhaust gas after-treatment.
3. Modeling, optimization and control for hybrid and electrified vehicles and their components (renewable energy storage systems, electric motors, power electronics, transmissions).
4. Connected and automated vehicle control.
5. Thermal management for next generation vehicles.
6. Applications of intelligent technologies to powertrains for supervision, energy management, diagnostics (telematics services, remote sensing).
7. Battery management, energy control and smart grid.
8. Intelligent control and energy markets design in smart grid.






E-COSM 2021 program will comprise several types of presentations

Regular Papers

Papers are invited in the form of regular manuscripts (limited to 6 pages; maximum two extra pages are allowed with 60 USD per extra page). Note that short manuscripts are not considered.

Invited Session Proposals

Proposals are sought from those wishing to organize an invited session. An invited session consists of 6 papers, which present a unifying theme from a diversity of viewpoints. The invited session papers have to be submitted separately by the respective authors, which will be individually reviewed along with the proposal itself.

Invited Session Papers

The six invited session papers must be submitted separately by their authors using the session code provided by the invited session organizer. Invited papers will undergo a review process similar to that of contributed papers. Invited papers must be submitted through the PaperPlaza Conference Manuscript Management System and must be formatted in accordance with the submission policy.

Discussion Papers

To foster active participation, we invite submissions of discussion papers, which should be at most 4 pages in length and formatted using standard conference format.

Industrial Papers

IFAC aims at increasing the industrial participation at all IFAC conferences. In this conference, we strongly encourage and solicit submissions of industrial papers, either regular or invited. Typically it is an extended abstract of 2 pages, however longer contributions are welcome.

Young Author Award

The IFAC Young Author Award will be granted to a young researcher at E-COSM 2021. Eligible candidates are authors of high-quality papers, younger than 30 at the time of event, and who will present the paper during the conference.
For details, see the event website.

Important Dates

Prospective authors are requested to submit their contributions in IFAC Conference Manuscript Management System https://ifac.papercept.net
The templates for manuscripts are available at the website and important dates for submission are as below

Submission open:	February 1, 2021
Invited for proposal papers:	March 05, 2021
Deadline for initial submission:	~~March 12, 2021~~ Extened to April 12, 2021
Acceptance notification:	~~May 1, 2021~~ Extened to June 1, 2021
Final papers:	~~May 31, 2021~~ Extened to June 30, 2021

Registration Fee

Advance full registration:	340USD
Later full registration:	380USD
Advance student registration:	240USD
Later student registration:	280USD
Virtual congress participation registeration (Advance):	150USD
Virtual congress participation registeration (Later):	180USD

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Call for Challengers

Benchmark Problem of Optimal Control of HEVs under V2X Connected Environment

Student Competition of *Benchmark Challenging* to powertrain control of connected vehicles will be organized. This competition is organized for students and younger researchers, which is as a part of conference program of IFAC E-COSM 2021. A poster session will be organized at the conference. Challengers might submit their challenging results with a discussion paper or a regular paper. The schedule of the problem is:

Open the problem:	August 1st, 2020
Submission of results:	May 31st, 2021
Final results submission:	June 30th, 2021

For downloading the simulator and sample data of driving route, all challengers must previously regist by sending challengers information to Dr. Fuguo Xu, Sophia University, Tokyo, Japan. fuguoxu@sophia.ac.jp. For detailed problem formulation see http://shenlab.jp/ecosm2021/program/benchmark-challenge.html.

Further information please visit: http://shenlab.jp/ecosm2021/index.html
Please contact us at: ecosm_ifac@ml.sophia.ac.jp

